ATOM COUNT CARD

SODIUM BICARBONATE

$NaHCO_3$

A. PRODUCT IDENTITY



Product: Relish All Purpose Baking Soda, Net Wt. 50 g
(Seanluc Inc.)

Compound: sodium bicarbonate, $NaHCO_3$
(ionic, so counted in formula units)

Why it's in the product: it is the leavening agent, releasing CO_2 gas that makes baked goods rise
(the pack also lists cleaning and deodorizing uses)

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B. BASE QUANTITATIVE DATA

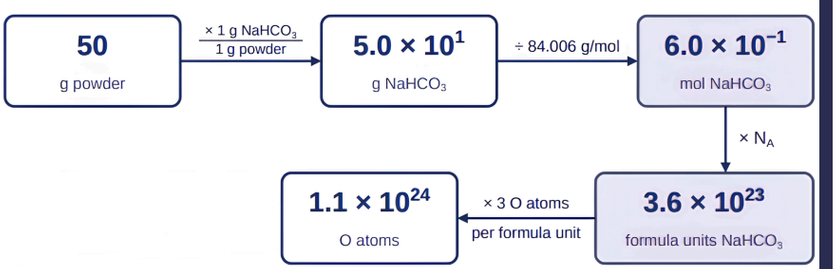
Quantity	Value	Source	Precision
Net content	50 g	Product label	**2 s.f. (limits)**
Purity assumed	100 % $NaHCO_3$	Ingredient list	Assumption
Serving size	½ tsp (2.4 g)	Product label	2 s.f.
Servings per container	about 21	Product label	Approximate
Atomic masses Na, H, C, O	22.990, 1.008, 12.011, 15.999 g/mol	CIAAW (2024)	Not limiting
Avogadro constant, N_A	$6.022\ 140\ 76 \times 10^{23}$ mol^{-1}	BIPM (2019)	Exact
Formula $NaHCO_3$	subscripts 1, 1, 3	PubChem (n.d.)	Exact counts

Assumption: the only ingredient is sodium bicarbonate, so the 50 g is treated as 100% $NaHCO_3$.

C. MOLAR MASS BUILD-UP

Element	Atoms	g/mol each	g/mol	% by mass
Na	1	22.990	22.990	27.367
H	1	1.008	1.008	1.200
C	1	12.011	12.011	14.298
O	3	15.999	47.997	57.135
$NaHCO_3$	**6**		**84.006**	**100.000**

D. THE MOLE MAP



E. MOLE CONVERTION CALCULATIONS

1 **Molar mass of $NaHCO_3$**

$1(22.990) + 1(1.008) + 1(12.011) + 3(15.999) =$ **84.006 g/mol**

2 **Label bridge: package → mass of $NaHCO_3$**

$50\ \cancel{\text{g powder}} \times \frac{1\ \text{g NaHCO}_3}{1\ \cancel{\text{g powder}}} = 50\ \text{g} \rightarrow$ **5.0×10^1 g $NaHCO_3$**

3 **Mass → moles**

$50\ \cancel{\text{g}} \times \frac{1\ \text{mol}}{84.006\ \cancel{\text{g}}} = 0.595196\ \text{mol} \rightarrow$ **6.0×10^{-1} mol $NaHCO_3$**

4 **Moles → formula units**

$0.595196\ \cancel{\text{mol}} \times \frac{6.022\ 140\ 76 \times 10^{23}\ \text{formula units}}{1\ \cancel{\text{mol}}} = 3.58435 \times 10^{23}$

→ **3.6×10^{23} formula units $NaHCO_3$**

5 **Atoms of one element inside the formula (oxygen)**

$3.58435 \times 10^{23}\ \cancel{\text{formula units}} \times \frac{3\ \text{O atoms}}{1\ \cancel{\text{formula unit}}} = 1.0753 \times 10^{24}$

→ **1.1×10^{24} O atoms**

6 **Percent composition + two-route check (sodium)**

$\%\text{Na} = \frac{22.990\ \text{g/mol}}{84.006\ \text{g/mol}} \times 100\ \% =$ **27.367 %**

Route A: $50\ \text{g} \times 0.27367 = 13.684\ \text{g}$

Route B: $0.595196\ \cancel{\text{mol NaHCO}_3} \times \frac{1\ \cancel{\text{mol Na}}}{1\ \cancel{\text{mol NaHCO}_3}} \times \frac{22.990\ \text{g Na}}{1\ \cancel{\text{mol Na}}} = 13.684\ \text{g}$

Both routes → **1.4×10^1 g Na** ✓ agree

7 **Reverse: from % composition to the formula**

In 100 g: Na $27.367 \div 22.990 = 1.1904$ mol

H $1.1999 \div 1.008 = 1.1904$ mol; C $14.298 \div 12.011 = 1.1904$ mol

O $57.135 \div 15.999 = 3.5712$ mol

$\div 1.1904 \rightarrow$ Na 1.000 : H 1.000 : C 1.000 : O 3.000

→ **$NaHCO_3$** (empirical = molecular, n = 1)

8 **Reverse: how many packs make one mole?**

$1.00\ \cancel{\text{mol}} \times \frac{84.006\ \cancel{\text{g}}}{1\ \cancel{\text{mol}}} \times \frac{1\ \text{pack}}{50\ \cancel{\text{g}}} = 1.6801\ \text{packs} \rightarrow$ **1.7 packs**

F. SCALE-UP FROM ONE PACK TO A WHOLE CLASS

1. ONE PACK
50 g of $NaHCO_3$
6.0×10^{-1} mol
3.6×10^{23} formula units

2. ONE CLASS
Assume 40 students × 1 pack
2.4×10^1 mol
= 23.808 mol; 2.0×10^3 g

3. ONE YEAR
Assume 1 pack per month (12)
7.1 mol
= 7.1423 mol; 6.0×10^2 g

4. REVERSE
One mole of $NaHCO_3$
1.7 packs
= 84.006 g of powder

G. SIGNIFICANT FIGURE CHECK

Calculator value: $3.584\ 351\ 686 \times 10^{23}$ formula units
Reported value: 3.6×10^{23} formula units $NaHCO_3$
Limits: 50 g (2 s.f.)
Exact: N_A, subscripts, student and pack counts
"50 g" has an ambiguous zero, so it is read as 2 s.f. *(5.0×10^1 g).*

ONE-NUMBER TAKEAWAY

"One 50 g pack holds about 3.6×10^{23} $NaHCO_3$ formula units, yet about 1.7 packs make a whole mole, so a mole is an enormous count that still weighs only 84 g."

H. REFERENCES

- Bureau International des Poids et Mesures. (2019). The International System of Units (SI) (9th ed.). https://www.bipm.org/en/publications/si-brochure
- Commission on Isotopic Abundances and Atomic Weights. (2024). Abridged standard atomic weights. International Union of Pure and Applied Chemistry. https://www.ciaaw.org/abridged-atomic-weights.htm
- National Center for Biotechnology Information. (n.d.). PubChem compound summary for CID 516892, sodium bicarbonate. Retrieved November 7, 2026, from https://pubchem.ncbi.nlm.nih.gov/compound/516892